LEARNING MODULE

27

Portfolio Return and Risk

A book's weights drift between trades, every trade costs money, and its risk splits among its funds in one way that adds up.

VOLUME V · Portfolio Optimization

BUILDS ON LM1 · LM3 · LM7 · LM8 · LM9 · LM10 · LM11 · LM12 · LM16 · LM19 · LM20 · LM22 · LM23

LEADS TO LM28 · LM29 · LM30 · LM31 · LM32 · LM33 · LM34 · LM35

TIME ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** compute a book's return from the weights set at the previous close, and track how buy-and-hold weights drift
☐	**2** derive the wealth recursion with trades, turnover and costs, and solve it for the post-trade wealth
☐	**3** explain why E5 needs an estimation frame and a wealth frame, and why its daily correlations can mislead
☐	**4** compute a portfolio's mean, variance and volatility from $\boldsymbol{\mu}$ and Σ, and measure its diversification
☐	**5** derive each asset's covariance and beta with the portfolio, its risk contributions, and Euler's identity
☐	**6** choose a meaningful split of portfolio risk, and interpret a negative or unstable contribution
☐	**7** split a portfolio's variance into systematic and residual parts with market betas
☐	**8** compute risk contributions and the wealth recursion in NumPy, and check them against their identities

LEARNING MODULE OVERVIEW

- A book earns its funds' returns weighted at the previous close, and between trades its weights drift: an equal-weight E5 book bought on 4 March 2019 and held had 48.6% in 0050 by 8 October 2026.
- Costs come out of wealth, so the post-trade wealth solves a self-financing equation, (27.2). A trade costs about 0.285% of its one-way turnover, plus the tax on taxed sales.
- R1's largest monthly reset, on 1 April 2026, traded 8.06% of the book and cost 0.031% of it: NT$3,100 on NT$10 million.
- Daily joint statistics of E5 use the 2,362 dates on which every fund has a one-session return; wealth uses all 2,368 sessions, including 0050's halt of June 2025.
- The funds close at 13:30 Taipei time, after the previous New York session, so daily correlations can mislead. Of five pairs, one daily–monthly gap reaches two standard errors: 00679B–00642U's, −0.18 daily against −0.36 monthly.
- Euler's identity splits a book's volatility into contributions that add up: equal weights put 45.9% of the risk in 00642U and 2.6% in 00679B.
- A negative contribution is a hedge only at the margin; 00679B's was negative in 22 of R1's 92 trailing windows.
- With market betas, 83.4% of an equal-weight U28 book's variance is systematic; most of the residual part comes from residual covariances.

1 INTRODUCTION

LM7 priced two funds at fixed weights, and LM8, LM10, LM16 and LM23 gave a book's return, its variance and the market model. This module adds time and attribution. Weights drift between trades, trades cost money out of wealth, five funds need two sets of dates, and the risk splits among them in one meaningful way.

MOTIVATING CASE **Which fund carries the risk?**

A reader holds LM7's two funds, 0050 and 00679B, and adds three: 00646 (the S&P 500), 00635U (gold futures) and 00642U (crude-oil futures). She puts a fifth of her money in each. On the 2,362 joint dates from January 2017 to October 2026, their volatilities ran from 13.49% to 37.43% a year. Yet the mix, reset to a fifth each day, had a volatility of 11.49% on the same dates.

Which fund carried that risk? 00642U held a fifth of the money and 45.9% of the risk; 00679B held a fifth and 2.6%. Left alone, the weights drift. Bought on 4 March 2019 and held, 0050 grew to 48.6% of the book by 8 October 2026.

Resetting the weights each month costs money. The largest of 91 monthly resets, on 1 April 2026, traded 8.06% of the book and cost 0.031% of its value. How should the risk be split, and what does each trade cost?

Exhibit 1 lists the five funds and the risk shares of two mixes, R1 and R2 (Section 3).

Exhibit 1: Five funds and two mixes

Fund	Index tracked	Volatility	R1 weight	R1 risk share	R2 weight	R2 risk share
0050	FTSE TWSE Taiwan 50	20.10%	20%	21.2%	40%	63.5%
00646	S&P 500	16.12%	20%	17.7%	20%	20.4%
00679B	ICE US Treasury 20+ Year	13.49%	20%	2.6%	30%	10.6%
00635U	S&P GSCI Gold Excess Return	16.60%	20%	12.6%	10%	5.5%
00642U	S&P GSCI Crude Oil Enhanced Excess Return	37.43%	20%	45.9%	0%	0%
The book's volatility				11.49%		11.17%
Weighted average volatility				20.75%		16.97%
Their ratio, (27.7)				1.81		1.52

Note: daily total returns on the 2,362 joint dates, divisor T, $A = 242.951$; weights reset daily; risk shares $w_i\beta_{i,p}$ by (27.6). R2's weights are *constructed*. Descriptive. Indices: Yuanta's fund pages. Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `TaiwanStockSplitPrice`, retrieved 24 October 2026. Computations: `data/lm27_numbers.py`.

Sections 2 to 4 cover returns, trades and E5's dates; Sections 5 to 8 measure and split the risk; Section 9 does it in Python.

NOTE **Notation for this module**

E5's $N = 5$ funds, indexed by i and j, come in the order 0050, 00646, 00679B, 00635U, 00642U; U28 has $N = 28$. Local:

- $R_{p,t}$, $R_{i,t}$: the book's and fund i's simple returns; σ_i, s_i: fund i's volatility in a model and in sample; $\rho_{i,p}$: the correlation of R_i with R_p.
- $\text{cost}(\mathbf{x})$: the cost of trades $\mathbf{x}$ in money; positions in money are $W_t^{\text{pre}}\mathbf{w}_t^{\text{drift}}$, since LM8's $\mathbf{h}$ counts units.
- $R_{\text{mkt},t}$, σ^2_{mkt}: the TAIEX price return and its variance; α_i, β_i, $\boldsymbol{\beta}$: market-model intercepts and betas (LM23); $\varepsilon_{i,t}$, Σ_{res}: the errors and their $N \times N$ covariance matrix, in sample that of the residuals (LM23's Σ_ε is a covariance across dates).
- h: each Deep Dive's function, defined in its first sentence; $a > 0$: a scalar in Deep Dive 2.

Plug-in values at $\hat{\Sigma}$ carry no hats ($\beta_{i,p}$, MRC_i, RC_i), nor do Section 8's OLS betas and sample moments (β_i, σ^2_{mkt}, Σ_{res}). Moments divide by T.

NOTE **Research hygiene**

Every matrix and risk share describes E5's daily, weekly or monthly returns (Section 4) or U28's sector price indices: **descriptive**. R1's shares in 92 trailing windows are a **stability check**. The trades, **fixed in advance** on 24 October 2026, show the accounting and judge no rule. Versions: R1, R2 and a buy-and-hold book. R2's weights and Problems 1–4 and 10 are **constructed**. Nothing is tested on later data (LM30, LM33, LM34).

2 THE RETURN OF A BOOK

☐ compute a book's return from the weights set at the previous close, and track how buy-and-hold weights drift

A book's return uses the weights held at the previous close, as an index's does (LM12). LM8's (8.3) and LM1's (1.13) are that sum for weights adding up to 1. A book may also hold cash, or borrow.

DEFINITION **A book's return and the drift of its weights**

Let $\mathbf{w}_{t-1}$ hold each fund's value over wealth after the close of date $t-1$, and let cash earn r_f a year. The book's simple return on date t, and its weights after that day's returns and before any trade, are

$$R_{p,t} = \mathbf{w}_{t-1}^{\top}\mathbf{R}_t + \left(1 - \mathbf{1}^{\top}\mathbf{w}_{t-1}\right)\frac{r_f}{A_{\mathrm{w}}}, \qquad w_{t,i}^{\mathrm{drift}} = \frac{w_{t-1,i}\left(1+R_{i,t}\right)}{1+R_{p,t}}. \tag{27.1}$$

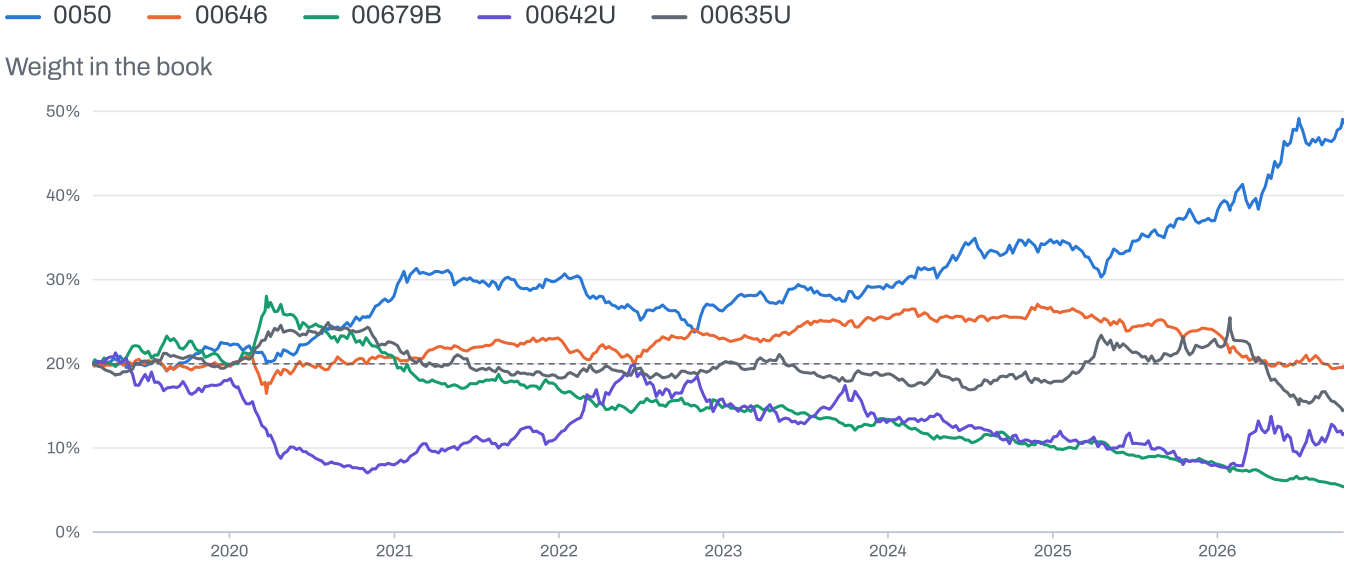
A negative cash balance, $\mathbf{1}^{\top}\mathbf{w}_{t-1} > 1$, pays LM31's financing rate in place of r_f. The drift needs $1 + R_{p,t} > 0$; otherwise the book is ruined on date t.

Fund i's holding grows from $W_{t-1}w_{t-1,i}$ to $W_{t-1}w_{t-1,i}(1+R_{i,t})$, and cash by the factor $1 + r_f/A_{\mathrm{w}}$. Their sum is the new wealth, $W_{t-1}(1+R_{p,t})$, and each holding over it is the drifted weight. For weights summing to 1 the cash term vanishes, and (27.1) is (8.3).

Between trades (27.1) applies daily, so a fund that beats the book gains weight. A book reset at every close is LM8's $X\mathbf{w}$, (8.12). A book that never trades keeps its holdings, (8.13), and drifts, as does a cap-weighted index between changes of its members or share counts (LM12). In LM9's buy-and-hold mix of two sectors, Semiconductor's weight drifted from 0.50 to 0.858.

Exhibit 2 shows the drift of an E5 book bought at a fifth each at the close of 4 March 2019. On 8 October 2026 it had 48.619%, 19.734%, 5.422%, 14.513% and 11.712% in the five funds. 0050's weight ran from 19.1% (15 August 2019) to 49.1% (1 July 2026). 00642U's fell from 21.3% (23 April 2019) to 7.0% (2 November 2020), and 00679B's peaked at 28.0% on 23 March 2020.

Exhibit 2: Weights of a buy-and-hold E5 book, 2019–2026



Note: wealth frame; no trade after the purchase, whose cost scales every holding alike. Plotted: each ISO week's last session, the purchase date and every fund's extremes, 402 dates. 00635U is grey; dashed, 20%. Descriptive. Source: as Exhibit 1.

EXAMPLE 1

Six sessions across 0050's halt

0050 stopped trading after 10 June 2025 for its 4-for-1 split and resumed on 18 June. Over those six sessions the five funds returned +0.864%, −1.689%, −0.792%, +2.226% and +12.816%. What did a book earn that held a fifth of each from the close of 10 June?

SOLUTION

By (27.1) applied once to the span, the book earns the funds' average, 2.685%: across funds, simple returns add (LM3). The average log return, 0.02524, gives $e^{0.02524} - 1 = 2.56\%$, too low, as (1.14) says of a long-only book.

Across time, log returns add. R1 itself, with its weights drifted since its trade of 2 June, earned 2.73%. Its log return is the sum of its six daily log returns, (1.12).

The drift needs $1 + R_{p,t} > 0$, the domain of $\ln(1 + R_{p,t})$. R1 and R2, long-only and fully invested, never leave it, since every fund's return exceeds −100%; LM35 shows a book that can.

PITFALL **Averaging log returns across funds**

For a long-only book the average of its funds' log returns understates its log return, (1.14). Average simple returns across funds and add log returns across days.

導讀 **權重會漂移，報酬要分兩個方向加**

帳戶當天的報酬，是前一天收盤時的權重（weights）乘上各基金當天的報酬再加總。不交易的時候，漲得比整個帳戶多的基金權重變大，落後的變小，這就是權重漂移（drift）。交易回到目標權重叫做再平衡（rebalancing），每天收盤都再平衡，權重就固定不變。完全不交易叫做買進持有（buy-and-hold）；按市值加權的指數在成分股與股數不變時就是買進持有。跨資產彙總要用簡單報酬（simple return），跨時間彙總要用對數報酬（log return）。把各基金的對數報酬平均，會低估只作多帳戶的報酬。

KNOWLEDGE CHECK 1

In Example 1, what is 00642U's weight at the close of 18 June, if the book does not trade?

ANSWER

The book grows by 1.02685, the average of the five growth factors. By (27.1) over the span, 00642U's weight is $0.2 \times 1.12816/1.02685 = 21.97\%$.

3 TRADES, TURNOVER AND COSTS

- ☐ derive the wealth recursion with trades, turnover and costs, and solve it for the post-trade wealth

The through case, fixed on 24 October 2026 before any result was computed, runs eight rules on E5. **R1** holds a fifth of each fund; **R2** holds a *constructed* mix of 40% 0050, 20% 00646, 30% 00679B and 10% 00635U. R3 to R8 are defined in LM28 to LM31, and every rule trades by (27.2).

A rule reads its window at a month's last close, known only after 13:30 (Research hygiene). It trades at the next close on which every fund trades, from $\mathbf{w}_t^{\text{drift}}$ to target exposures $\mathbf{f}_t$. The costs come out of wealth, so the book is **self-financing**.

DEFINITION **The wealth recursion**

With W_{t-1} the wealth after date $t-1$'s trade, the wealth before and after date t's trade and the cost of trades $\mathbf{x}$ in money are

$$\begin{aligned} W_t^{\text{pre}} &= W_{t-1}(1+R_{p,t}), \\ W_t &= W_t^{\text{pre}} \\ &\quad - \operatorname{cost}\left(W_t\mathbf{f}_t - W_t^{\text{pre}}\mathbf{w}_t^{\text{drift}}\right), \\ \operatorname{cost}(\mathbf{x}) &= 0.1425\%\,\|\mathbf{x}\|_1 \\ &\quad + \textstyle\sum_i \operatorname{tax}_i \max(0,-x_i). \end{aligned} \tag{27.2}$$

After a trade $\mathbf{w}_t = \mathbf{f}_t$. On a date without a trade, $W_t = W_t^{\text{pre}}$ and $\mathbf{w}_t = \mathbf{w}_t^{\text{drift}}$.

The trade depends on the post-trade wealth, so W_t sits on both sides. The commission is 0.1425% of the value traded on each side, and the tax 0.1% of every sale except 00679B's, a bond ETF exempt through 2026. These are Taiwan's posted commission (TWSE's Operating Rules, Art. 94; TPEx's Business Rules, Art. 64, which its ETF rules apply) and transaction tax (Securities Transaction Tax Act, Arts. 2 and 2-1). LM34 discusses them. No slippage is charged.

RESULT **One post-trade wealth**

If the factor $(0.1425\% + \max_i \text{tax}_i)\,\|\mathbf{f}_t\|_1$ is below 1, then (27.2) has exactly one solution W_t. If selling the whole book would also cost less than W_t^{pre}, then $0 < W_t \le W_t^{\text{pre}}$ (Deep Dive 1).

For E5's rules the factor is below 0.4%: 0.2425% for a fully invested long-only book. To size a trade, replace W_t by W_t^{pre} inside the cost:

$$\begin{aligned} \text{TO}_t &= \tfrac{1}{2}\|\mathbf{f}_t - \mathbf{w}_t^{\text{drift}}\|_1, \\ \frac{\text{cost}}{W_t^{\text{pre}}} &\approx 0.285\%\,\text{TO}_t \\ &\quad + \textstyle\sum_i \text{tax}_i \max\left(0,\, w_{t,i}^{\text{drift}} - f_{t,i}\right), \\ |\text{error}| &\le \frac{\text{cost}}{W_t^{\text{pre}}}\left(0.1425\% + \max_i \text{tax}_i\right)\|\mathbf{f}_t\|_1. \end{aligned} \tag{27.3}$$

The one-way turnover TO_t is half LM8's 1-norm, since for weights summing to 1 before and after a trade the buys equal the sales. A trade then costs 0.285% of its turnover with no taxed sale, and up to 0.385% when every sale is taxed. Problem 8 derives the bound.

DEEP DIVE **Deep Dive 1: one post-trade wealth** optional · click to expand

Here $h(W) = W + \text{cost}(W\mathbf{f}_t - W_t^{\text{pre}}\mathbf{w}_t^{\text{drift}})$, so (27.2) reads $h(W_t) = W_t^{\text{pre}}$. The cost is Lipschitz in $\mathbf{x}$ with constant $0.1425\% + \max_i \text{tax}_i$: a change in $\mathbf{x}$ moves it by at most that constant times the change's 1-norm. So a move of W moves it by at most the result's factor times as much. Hence h is continuous, with slope at least 1 minus that factor, and takes the value W_t^{pre} once.

The root is at most W_t^{pre}, since a cost is never negative. It is positive when $h(0)$, the cost of selling the whole book, is below W_t^{pre}, as for every E5 rule. The iteration $W \mapsto W_t^{\text{pre}} - \text{cost}(\cdot)$ multiplies the distance to the root by at most the same factor at each step, so it converges geometrically.

EXAMPLE 2

R1's first purchase

R1 starts with cash of 1 and buys a fifth of each fund at the close of 4 March 2019. Find the wealth after the trade, the cost and the turnover.

SOLUTION

Every trade is a buy, so no tax is due, and the trades sum to W_t in money. By (27.2), $W_t = 1 - 0.001425\,W_t$, so $W_t = 1/1.001425 = 0.998577$. The cost is 0.1423% of the pre-trade wealth of 1, and $\text{TO}_t = \tfrac{1}{2}\|\mathbf{f}_t\|_1 = 0.5$, half the buys, since cash pays for them. R2's first purchase is the same.

EXAMPLE 3

R1's trade of 1 April 2026

After the signal of 31 March 2026, R1's drifted weights at the close of 1 April were 17.954%, 18.611%, 18.697%, 16.680% and 28.058%. It traded back to a fifth each. Find the turnover and the cost, by (27.3) and exactly.

SOLUTION

The turnover is half the sum of the five gaps to 20%: $\mathrm{TO}_t = 8.058\%$. Only 00642U is sold, by 8.058 points, so (27.3) gives, as a share of the pre-trade wealth,

$$0.285\% \times 8.058\% + 0.1\% \times 8.058\% = 0.03102\%.$$

The exact root of (27.2) gives 0.03100%: NT$3,100 on a book of NT$10,000,000 (Exhibit 3). It was the largest of R1's 91 rebalances by turnover.

Exhibit 3: R1's trade of 1 April 2026 on a book of NT$10 million

Fund	Before the trade	Trade	After the trade	Commission	Tax
0050	1,795,428	+203,952	1,999,380	291	0
00646	1,861,080	+138,300	1,999,380	197	0
00679B	1,869,692	+129,688	1,999,380	185	0
00635U	1,668,005	+331,375	1,999,380	472	0
00642U	2,805,795	−806,415	1,999,380	1,149	806
Total	10,000,000	−3,100	9,996,900	2,294	806

Note: NT$, whole dollars; every column sums as printed. The through case's R1, signal 31 March 2026; the book's size is hypothetical, since (27.2) is scale-free. One-way turnover 8.06%. Source: as Exhibit 1.

Cash earns an illustrative $r_f = 1.5\%$ a year, r_f / A_{w} a session; R1 and R2 hold none after the first purchase. LM31 sets R8's exposure, which may exceed 1, and the rate at which R8 borrows. No final sale is charged on 8 October 2026.

PITFALL **Turnover counted on both sides**

Adding the buys and the sales doubles the one-way turnover, which is half the 1-norm. The commission falls on both sides, so (27.3)'s rate is $2 \times 0.1425\%$ of TO_t; the tax falls on sales only.

導讀 **交易成本從財富裡扣**

再平衡要買賣，每一筆都付手續費（commission），賣出還要付證券交易稅（securities transaction tax）。這些交易成本（transaction cost）直接從帳戶財富扣掉，不另外放錢進來，所以叫做自我融資（self-financing）。麻煩在於交易後的財富決定要買多少，買多少又決定成本，所以它同時出現在等式兩邊；只要成本率乘上總曝險（gross exposure）小於 1，就只有一個解，可以反覆代入求出。週轉率（turnover）是單邊的，等於權重差的 L_1 範數（1-norm）的一半；把買進和賣出加總會算成兩倍。

KNOWLEDGE CHECK 2

Switching NT$1,000,000 from 00679B to 0050 costs about NT$2,850. Why does the reverse switch cost about NT$3,850?

ANSWER

Each switch pays 0.1425% of commission on each side, NT$2,850. The reverse switch also pays the 0.1% tax on 0050's sale, NT$1,000; a sale of 00679B pays none. Both use (27.3), ignoring the cost on the cost.

4 E5: TWO FRAMES AND ASYNCHRONOUS CLOSES

- [ ] explain why E5 needs an estimation frame and a wealth frame, and why its daily correlations can mislead

E5's funds track the indices of Exhibit 1; those of 00635U and 00642U follow COMEX gold and NYMEX WTI crude-oil futures. Returns are total returns in New Taiwan dollars, $R_{i,t} = (P_t + D_t)/P_{t-1} - 1$, and on 0050's 4-for-1 split of 18 June 2025 the return is $(P_t + D_t)/(P_{t-1}/4) - 1$. The sample runs from the base close of 17 January 2017, 00679B's first, to 8 October 2026: 3,551 days, or 9.7221 years.

The funds were chosen in 2026 among funds still listed; LM34 shows what such a choice can hide. Joint statistics need every fund's return over the same session, but wealth must cover every session. The **estimation frame** holds the dates on which every fund has a one-session return: $T = 2{,}362$, and $A = 2{,}362 \times 365.25/3{,}551 = 242.951$ as LM10 counts a year. The **wealth frame** holds all 2,368 sessions, $A_{\mathrm{w}} = 243.569$; a halted fund earns 0 there until it trades again.

Every daily covariance, correlation, mean and risk share uses the estimation frame, "on the 2,362 joint dates". Weekly and monthly returns, and every book's return, drift, trade and wealth, use the wealth frame.

EXAMPLE 4

Two frames, one halt

Why do E5's frames differ by six dates, and what does that do to 00642U's growth factor?

SOLUTION

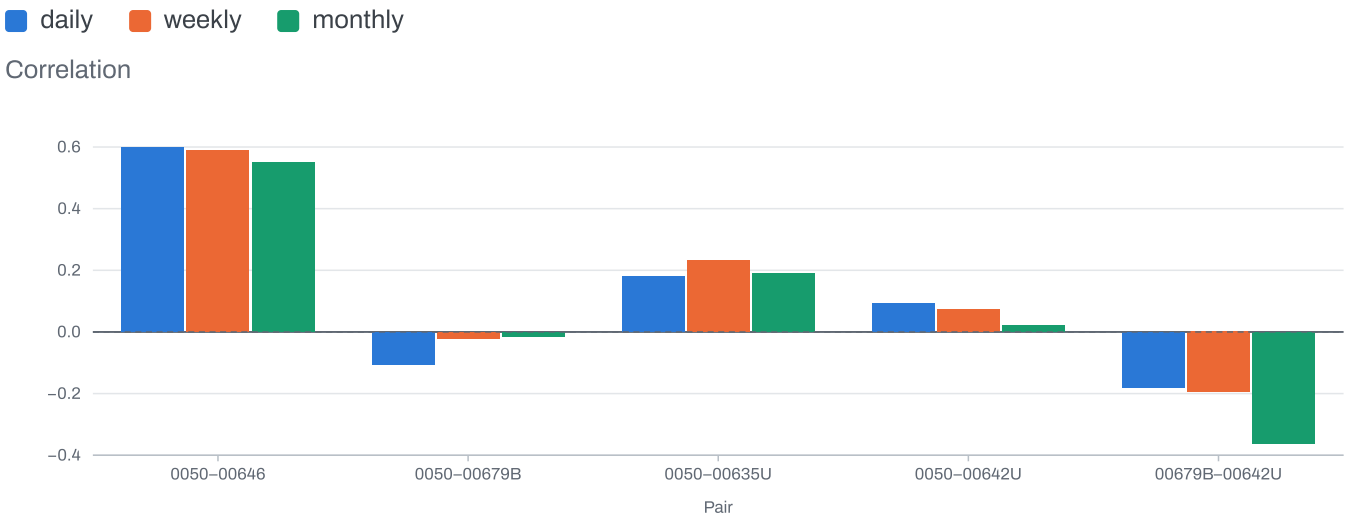
The six dates 11 to 18 June 2025 are in the wealth frame only. 0050 earns 0 on the first five and $4 \times 47.57/188.65 - 1 = 0.864\%$ on 18 June, from its close of 10 June. Lacking 0050's one-session returns, the estimation frame drops all six. So 00642U's growth factor is 1.593 over the wealth frame and 1.412 over the estimation frame, which misses its +12.816% of Example 1.

All five funds trade in Taipei, 00679B on TPEx and the others on TWSE. Both markets match orders from 09:00 to 13:30 (TWSE's Operating Rules, Art. 3; TPEx's Business Rules, Art. 40, which its ETF rules apply). Each closes with a call auction from 13:25, as its trading-system page says. New York's core session ends at 4:00 p.m. (NYSE): 04:00 Taipei time in US summer time, 05:00 otherwise.

So the 13:30 closes of 00646 and 00679B follow the previous New York session, and 0050 reacts to it too. TWSE's ETF overview page adds that a futures-trust ETF on a foreign futures index is valued on European and US futures, so the time difference is a risk. Across markets that close at different times, correlations are understated (Burns, Engle and Mezrich 1998). Here the closes coincide but reflect different sessions, so daily correlations can mislead either way.

Exhibit 4 compounds the wealth frame's returns within ISO weeks and calendar months, so each is a return from close to close. It drops the two weeks across 0050's halt, since 0050 has no close on 13 June 2025. Weeks are annualized as LM19 counts a year. The 500 weeks with a session span 9.6975 years, from the close of 20 January 2017 to that of 2 October 2026, so $A = 500 \times 365.25/3{,}542 = 51.56$; months use $A = 12$.

Exhibit 4: Correlations of E5's funds by frequency



Note: daily simple returns on the 2,362 joint dates, and weekly and monthly ones from the wealth frame (498 ISO weeks, 116 calendar months). The first and last week and month are dropped, complete or not, and so are the two weeks across 0050's halt. Near zero correlation, normal-theory standard errors are about 0.02, 0.045 and 0.09 (0050–00679B: 0.020, 0.045, 0.093). Plug-in ones are 1.4 to 2.4 times as large on daily data (0050–00679B: 0.040, 0.053, 0.095). Descriptive. Source: as Exhibit 1.

EXAMPLE 5

Correlations by frequency

Does a daily correlation in Exhibit 4 differ from the weekly or monthly one by more than chance?

SOLUTION

Daily, weekly and monthly, 0050 correlates −0.105, −0.022 and −0.015 with 00679B, 0.599, 0.588 and 0.549 with 00646, and 0.180, 0.231 and 0.191 with 00635U. With 00642U, 0050 correlates 0.092, 0.072 and 0.022. 00679B–00642U gives −0.18, −0.19 and −0.36.

The normal-theory errors are $(1-\hat{\rho}^2)/\sqrt{T}$; the plug-in ones of LM20's Problem 6 drop normality but need finite fourth moments, and both assume independent periods. So read 00642U's plug-in errors with care: its raw kurtosis (19.3), 26.6 daily and 12.3 monthly, is the largest (the others 11.2 to 21.0 and 4.3 to 5.7).

Of the five pairs, no daily–weekly gap reaches two weekly standard errors of either kind; the largest, 0.0833 for 0050–00679B, is 1.9 normal-theory ones. The daily figure is higher for 0050–00646, 0050–00642U and 00679B–00642U, and lower for the other two. Of the five pairs, one daily–monthly gap reaches two monthly standard errors: 00679B–00642U's 0.1807 (from the unrounded correlations), 2.2 normal-theory ones (0.0807) and 2.0 plug-in ones (0.0890). The data name no cause.

A daily correlation and a weekly or monthly one come from the same days, so their gap has its own standard error. Resampling whole weeks or months (LM22) gives about 0.046 for 0050–00679B's daily–weekly gap, 1.8 of them, and about 0.08 for 00679B–00642U's daily–monthly gap, which stays above two of them. Experiment 1 draws 20,000 resamples of each, weeks first, from `default_rng([1, 20261011])`, with Monte Carlo errors below 0.001.

Monthly plug-in errors reach 0.16. R1's volatility is 11.49%, 10.57% and 10.03% a year at the three frequencies, and 00642U's share 45.9%, 41.9% and 44.9%. The rules use daily $\hat{\Sigma}$.

導讀 **同一個收盤時間，不同的海外交易時段**

五檔指數股票型基金（ETF）都在台北 13:30 收盤，但 00646 和 00679B 的價格反映前一晚的紐約交易時段，0050 在同一個收盤也受那個時段影響。期貨型 ETF 依歐美期貨評價，時差本身就是風險。這種非同步（asynchronous）收盤會讓日報酬的相關係數（correlation coefficient）和週、月報酬的不同，而且方向不一定。比較不同頻率時要附上標準誤（standard error）：兩種標準誤都假設各期獨立，plug-in 標準誤還需要四階動差（fourth moment）有限，所以要同時看各基金的峰態（kurtosis）。

KNOWLEDGE CHECK 3

(a) Why does E5's 0050/00679B pair have 2,362 returns and LM7's 2,363? (b) Why can 0050's daily correlation with 00646 differ from the weekly one?

ANSWER

(a) The estimation frame drops 18 June 2025, whose return spans 0050's halt. LM7 measured the pair from each previous common close, keeping one return from 10 to 18 June. That sample gives $\hat{\rho} = -0.1056$ (LM8, recomputed here), against −0.105 on E5's 2,362.

(b) 00646's 13:30 price follows the previous New York session, while 0050 also moves with Taipei's own trading. News can enter the two daily returns on different days but usually in the same week.

5 PORTFOLIO MEAN AND VARIANCE

- [ ] compute a portfolio's mean, variance and volatility from $\boldsymbol{\mu}$ and Σ, and measure its diversification

Let $\mathbf{R}$ have mean $\boldsymbol{\mu}$ and covariance matrix Σ, and hold constant weights $\mathbf{w}$. The return $R_p = \mathbf{w}^\top \mathbf{R}$ is linear in $\mathbf{R}$, so its mean is $\mathbf{w}^\top \boldsymbol{\mu}$, and (16.4) with $M = \mathbf{w}^\top$ gives its variance:

$$\mu_p = \mathbf{w}^\top \boldsymbol{\mu}, \qquad \sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}, \qquad \text{a year: } A\mu_p \text{ and } \sqrt{A}\,\sigma_p. \tag{27.4}$$

In sample the same steps give (10.4), $s_p^2 = \mathbf{w}^\top \hat{\Sigma} \mathbf{w}$, exact for a book reset at every close. The annual forms are conventions on simple returns, since variances add only for log returns on uncorrelated periods, (19.7). Every matrix here is a plug-in estimate. Section 4 measures the standard errors of five of the daily correlations; LM30 measures the error of $\hat{\Sigma}$ and of R1's risk shares.

Exhibit 5: E5's means, volatilities and correlations

	0050	00646	00679B	00635U	00642U
Mean, a year	24.16%	14.87%	−0.97%	9.26%	10.53%
Volatility, a year	20.10%	16.12%	13.49%	16.60%	37.43%
0050	1	0.60	−0.11	0.18	0.09
00646	0.60	1	−0.08	0.09	0.20
00679B	−0.11	−0.08	1	0.18	−0.18
00635U	0.18	0.09	0.18	1	0.02
00642U	0.09	0.20	−0.18	0.02	1

Note: daily total returns on the 2,362 joint dates, divisor T, $A = 242.951$. Means are $A\bar{R}_i$ and volatilities $\sqrt{A}\,s_i$. The last five rows are the correlation matrix C of (10.3). Descriptive. Source: as Exhibit 1.

By (8.11), for $\mathbf{w} \geq \mathbf{0}$ the book's volatility is at most the weighted average of the funds' volatilities. Their ratio therefore measures diversification:

$$\frac{\mathbf{w}^{\top}\mathbf{s}}{s_p} \geq 1 \qquad (\mathbf{w} \geq \mathbf{0},\ s_p > 0), \tag{27.7}$$

with equality only when every correlation among the held assets is 1. Choueifaty and Coignard (2008, eq. (1)) name it the **diversification ratio**. For a long-only book that is not equivalent to a single asset, it exceeds 1.

EXAMPLE 6

R1 and R2: mean, variance and diversification

From Exhibit 5, find μ_p, s_p and the ratio (27.7) of R1 and R2 at the full-sample $\hat{\Sigma}$.

SOLUTION

R1's μ_p is the average of the five annual means, 11.57%, and its s_p is 11.49% (full sample, weights reset daily). By (10.6), the variance terms make 76.4% of s_p^2 and the covariance terms 23.6%, against 7.2% and 92.8% for LM10's U28 book. The weighted average volatility is 20.75%, so the ratio (27.7) is 1.81.

For R2,

$$\begin{aligned}\mu_p &= 0.4 \times 24.16\% + 0.2 \times 14.87\% \\ &\quad + 0.3 \times (-0.97\%) \\ &\quad + 0.1 \times 9.26\% = 13.27\%.\end{aligned}$$

Its volatility is 11.17% and its weighted average volatility 16.97%, a ratio of 1.52. LM10's equal-weight U28 book has 21.94% over 15.75%, a ratio of 1.393. Each book is long-only and not equivalent to a single asset, so each ratio exceeds 1, as Choueifaty and Coignard state.

KNOWLEDGE CHECK 4

The eigenvalues of $A\hat{\Sigma}$ run from $\lambda_5 = 118.05$ to $\lambda_1 = 1{,}429.89$ (%²). Between what bounds does (11.4) put R1's s_p^2?

ANSWER

R1's weights have $\|\mathbf{w}\|^2 = 5 \times 0.2^2 = 1/5$, so $s_p^2 = \frac{1}{5}\,\mathbf{u}^\top A\hat{\Sigma}\mathbf{u}$ with the unit vector $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. By (11.4), s_p^2 lies between $\lambda_5/5 = 23.61$ and $\lambda_1/5 = 285.98$ (%²). It is 132.09, the square of R1's full-sample 11.493%.

6 WHICH FUND CARRIES THE RISK

- ☐ derive each asset's covariance and beta with the portfolio, its risk contributions, and Euler's identity
- ☐ choose a meaningful split of portfolio risk, and interpret a negative or unstable contribution

By (16.2) and the bilinearity of covariance, $\mathrm{Cov}(R_i, R_p) = \sum_j \sigma_{ij} w_j$, entry i of $\Sigma\mathbf{w}$, as LM7 found for two assets. Dividing by σ_p^2 gives fund i's beta to the portfolio, the slope of its line on R_p by (12.3):

$$\begin{aligned}\mathrm{Cov}(R_i, R_p) &= (\Sigma\mathbf{w})_i, \\ \beta_{i,p} = \frac{(\Sigma\mathbf{w})_i}{\sigma_p^2}, &\qquad \sum_i w_i \beta_{i,p} = 1.\end{aligned} \tag{27.5}$$

The last identity holds because $\sum_i w_i(\Sigma\mathbf{w})_i = \mathbf{w}^\top\Sigma\mathbf{w} = \sigma_p^2$. Both ratios need $\sigma_p > 0$.

The **marginal risk contribution** MRC_i is the rate at which the book's volatility changes with fund i's weight, the other weights fixed. By (10.8), $\nabla\sigma_p^2 = 2\Sigma\mathbf{w}$, and for $\sigma_p > 0$ the chain rule gives $\partial\sigma_p/\partial w_i = (\Sigma\mathbf{w})_i/\sigma_p$. Multiplying by w_i gives the **risk contribution** RC_i:

$$\begin{gathered}\mathrm{MRC}_i = \frac{\partial\sigma_p}{\partial w_i} = \frac{(\Sigma\mathbf{w})_i}{\sigma_p},\\ \mathrm{RC}_i = w_i\,\mathrm{MRC}_i, \qquad \textstyle\sum_i \mathrm{RC}_i = \sigma_p,\\ \mathrm{RC}_i/\sigma_p = w_i\beta_{i,p}.\end{gathered} \tag{27.6}$$

The sum is **Euler's identity**, which holds because σ_p is homogeneous of degree 1 in $\mathbf{w}$: doubling every weight doubles the volatility (Deep Dive 2). Maillard, Roncalli and Teïletche (2010) define the contributions this way. The **risk share** $w_i\beta_{i,p}$ is RC_i over σ_p. LM10 warned that entry i of $2\hat{\Sigma}\mathbf{w}$ is a rate of change, not a share: divided by $2s_p$ it is MRC_i, and the share is $w_i\,\mathrm{MRC}_i/s_p$.

> **DEEP DIVE** **Deep Dive 2: Euler's theorem** optional · click to expand
>
> Here h is differentiable at $\mathbf{x}$, with $h(a\mathbf{x}) = a\,h(\mathbf{x})$ for every $a > 0$. Differentiating in a at $a = 1$ gives $\sum_i x_i\,\partial h/\partial x_i = h(\mathbf{x})$. The volatility $\sqrt{\mathbf{w}^\top\Sigma\mathbf{w}}$ is such a function wherever $\sigma_p > 0$, and not differentiable at a book of zero variance. The variance, homogeneous of degree 2, gives $\sum_i w_i\,\partial\sigma_p^2/\partial w_i = 2\sigma_p^2$, which by (10.8) reads $\sum_i w_i\,\mathrm{Cov}(R_i, R_p) = \sigma_p^2$. The working paper of Maillard et al. states the split this way (§2.1, p. 4, and its footnote).

EXAMPLE 7

R1's risk, split three ways

Split R1's full-sample volatility among the five funds by Euler's identity, and compare two other splits.

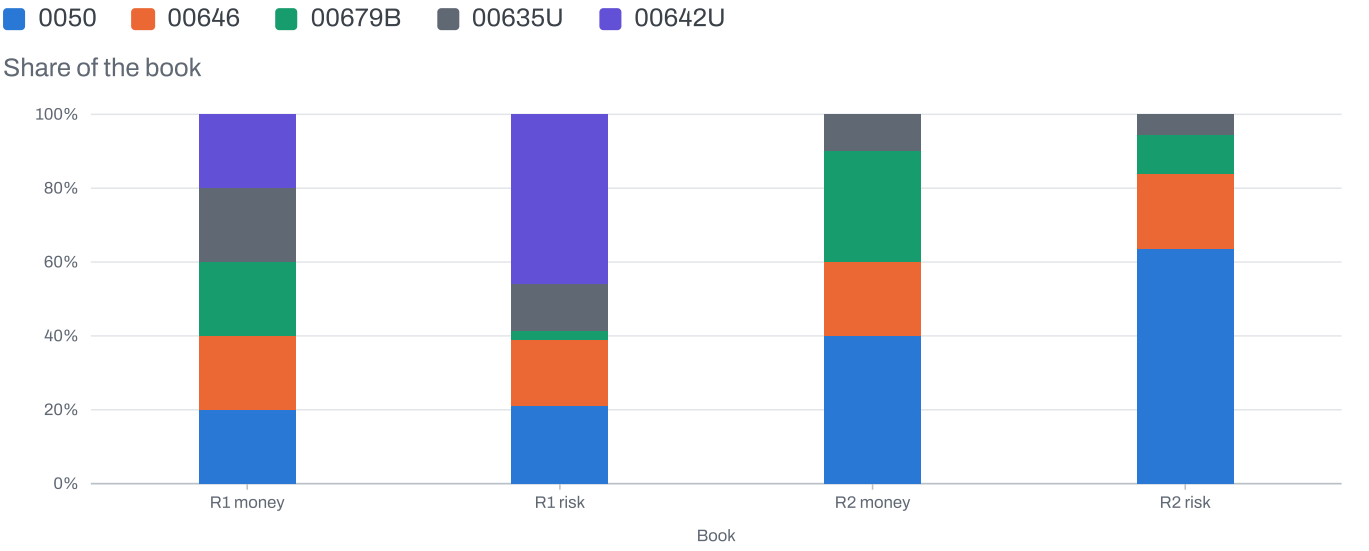
SOLUTION

At the full-sample $\hat{\Sigma}$ on the 2,362 joint dates, $A(\hat{\Sigma}\mathbf{w})_i$ is 139.80, 117.05, 17.22, 83.25 and 303.13 (%²). Dividing by $s_p^2 = 132.09$ (%²) gives $\beta_{i,p}$ of 1.058, 0.886, 0.130, 0.630 and 2.295, so the shares $w_i\beta_{i,p}$ are 21.2%, 17.7%, 2.6%, 12.6% and 45.9% (Exhibit 6). From unrounded inputs, MRC_i is 12.16%, 10.18%, 1.50%, 7.24% and 26.37%, and RC_i 2.43, 2.04, 0.30, 1.45 and 5.27 points, summing to 11.49.

Split	0050	00646	00679B	00635U	00642U	Sum
Euler, $w_i\beta_{i,p}$	21.2%	17.7%	2.6%	12.6%	45.9%	100.0%
Stand-alone, $w_i s_i / \mathbf{w}^\top \mathbf{s}$	19.4%	15.5%	13.0%	16.0%	36.1%	100.0%
Variance terms, $w_i^2 s_i^2 / s_p^2$	12.240%	7.873%	5.514%	8.340%	42.433%	76.400%

Only Euler's parts add up to the book's volatility, 11.49 points, and only they price a switch. The stand-alone parts $w_i s_i$ add up to the weighted average volatility, 20.75%, and the variance terms (from unrounded inputs) to 76.4% of the variance. The OLS slopes of $w_i R_{i,t}$ on $R_{p,t}$ equal the shares to about 10^{-16}, machine precision.

Exhibit 6: Money against risk in R1 and R2



Note: weights and risk shares $w_i\beta_{i,p}$ at the full-sample $\hat{\Sigma}$ on the 2,362 joint dates; each bar sums to 100%. 00635U is the grey segment; R2 holds no 00642U. Descriptive. Source: as Exhibit 1.

Euler's split is the meaningful one, for five reasons. It adds up to the book's volatility. Fund i's share of the variance is its variance term plus half of each covariance term it enters (Problem 7). Its MRCs price a switch: moving a small weight from fund j to fund i changes σ_p at the rate $\mathrm{MRC}_i - \mathrm{MRC}_j$. By (12.3), the share is the slope of $w_i R_{i,t}$ on $R_{p,t}$.

Last, under joint normality (16.8) makes $\mathbb{E}[w_i R_i \mid R_p]$ a line in R_p with slope $w_i\beta_{i,p}$. So in a large move of the book, fund i's expected fraction of the move is close to its share (Qian 2006). Qian extends that

reading to value at risk, which this book does not split by fund. Equal contributions as a target are LM31's risk parity. At LM28's minimum-variance portfolio every $\beta_{i,p}$ is 1, so risk shares equal weights.

EXAMPLE 8

A switch of five points

Move 5 points of R1 from 00642U to 00679B. Predict the change in s_p from the MRCs, and compute it exactly by (10.9).

SOLUTION

The linear prediction is $0.05\,(\mathrm{MRC}_{00679B} - \mathrm{MRC}_{00642U}) = -1.24$ points. With $\boldsymbol{\Delta}$ holding +0.05 and −0.05, (10.9) gives the new variance exactly, in $\%^2$:

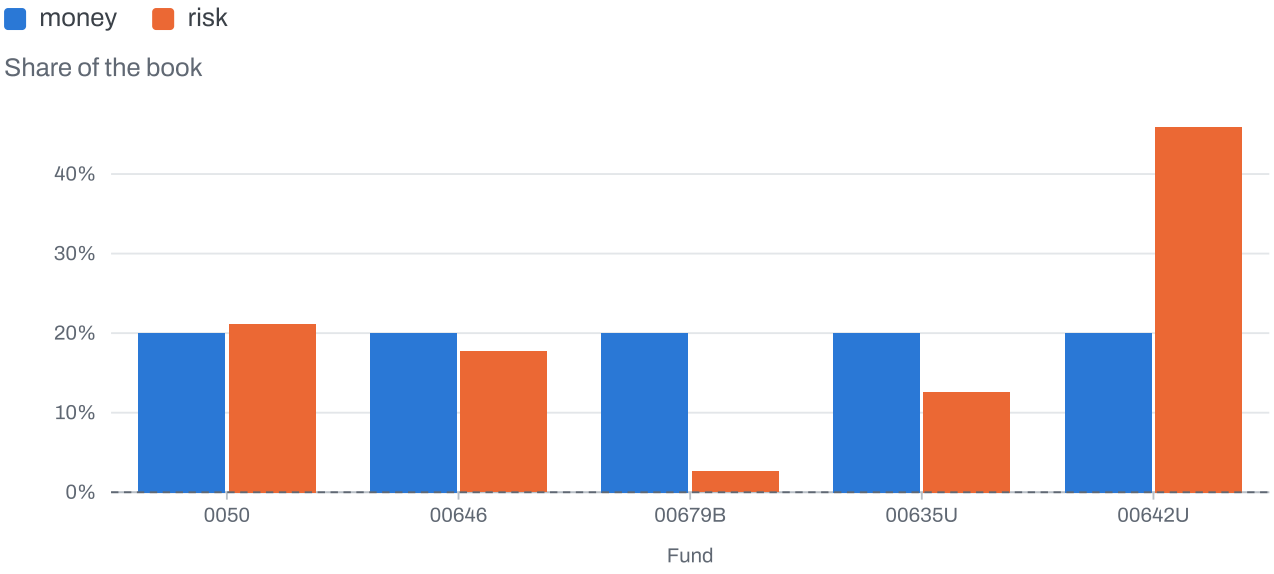
$$132.09 - 28.59 + 4.42 = 107.92.$$

So s_p falls from 11.49% to 10.39%, by 1.10 points. The quadratic term $\boldsymbol{\Delta}^{\top} A \hat{\Sigma} \boldsymbol{\Delta} = 4.42$, from unrounded inputs, is why the linear prediction overshoots.

The widget applies (27.5) and (27.6) to a fully invested book of the five funds, shorts included, under the daily, monthly or Section 7's window matrix. It applies (27.7) while no weight is negative.

Weights against risk shares

INTERACTIVE



Book's volatility, a year **11.49%**	Weighted average volatility **20.75%**	Ratio (27.7) **1.81**
Sum of the risk shares **100.0%**	0050: money, risk **20.0%, 21.2%**	00646: money, risk **20.0%, 17.7%**
00679B: money, risk **20.0%, 2.6%**	00635U: money, risk **20.0%, 12.6%**	00642U: money, risk **20.0%, 45.9%**

Descriptive: E5's total returns, annualized, divisor T: daily on the 2,362 joint dates or the window's 504, or monthly over 116 calendar months. 00642U takes the rest of the money. A risk share is $w_i(\Sigma w)_i/s_p^2$, so the shares always add up to 100%. A negative share of a fund held long means that more of it lowers the book's volatility at the margin.

導讀 誰承擔了風險

投資組合（portfolio）的波動率（volatility）不是各基金波動率的加總，但可以拆成加起來剛好等於它的幾份。邊際風險貢獻（marginal risk contribution）是某檔基金權重增加一點時整體波動率增加的速度，乘上權重就是風險貢獻（risk contribution）。波動率是一次齊性函數（homogeneous function of degree 1）：所有權重同乘兩倍，波動率也乘兩倍，所以由 Euler 定理（Euler's theorem），各基金的風險貢獻加總剛好等於整體波動率。等權重看起來很風險分散（diversification），但 00642U 只占五分之一的錢，卻承擔了 45.9% 的風險。

KNOWLEDGE CHECK 5

(a) In R2, 0050 holds 40% of the money and has $\beta_{i,p} = 1.587$. What is its risk share, and why do R2's five shares sum to 1? (b) Example 7 gives 00642U three parts of R1's risk: 36.1% stand-alone, a variance term of 42.433% and an Euler share of 45.9%. Which would you report, and why not the other two?

ANSWER

(a) 0050's share is $0.4 \times 1.587 = 63.5\%$. The shares sum to $\sum_i w_i \beta_{i,p} = \mathbf{w}^\top \Sigma \mathbf{w} / \sigma_p^2 = 1$, by (27.5).

(b) Report the Euler share, 45.9%. Only Euler's parts add up to the book's volatility, and only their MRCs price a switch (Example 8). The stand-alone parts add up to the weighted average volatility, 20.75%, and the variance terms to only 76.4% of the variance.

7 NEGATIVE AND MOVING CONTRIBUTIONS

- [ ] choose a meaningful split of portfolio risk, and interpret a negative or unstable contribution

A risk contribution is negative exactly when $w_i \operatorname{Cov}(R_i, R_p) < 0$, as for a long position in a fund that moves against the book. It lowers the book's volatility at the margin, not at any size: with the other weights fixed,

$$\begin{aligned} \frac{\partial \operatorname{MRC}_i}{\partial w_i} &= \frac{\sigma_i^2 - \beta_{i,p}^2 \sigma_p^2}{\sigma_p} \\ &= \frac{\sigma_i^2 \left(1 - \rho_{i,p}^2\right)}{\sigma_p} \geq 0 \end{aligned}$$

(Problem 5). So MRC_i never falls as w_i grows. It tends to $\sigma_i > 0$ as w_i grows without bound, so a hedge at the margin stops hedging at a large enough weight.

EXAMPLE 9

00679B as a hedge

In the window of 504 joint dates from 29 January 2018 to 27 February 2020, would Example 8's switch lower R1's volatility?

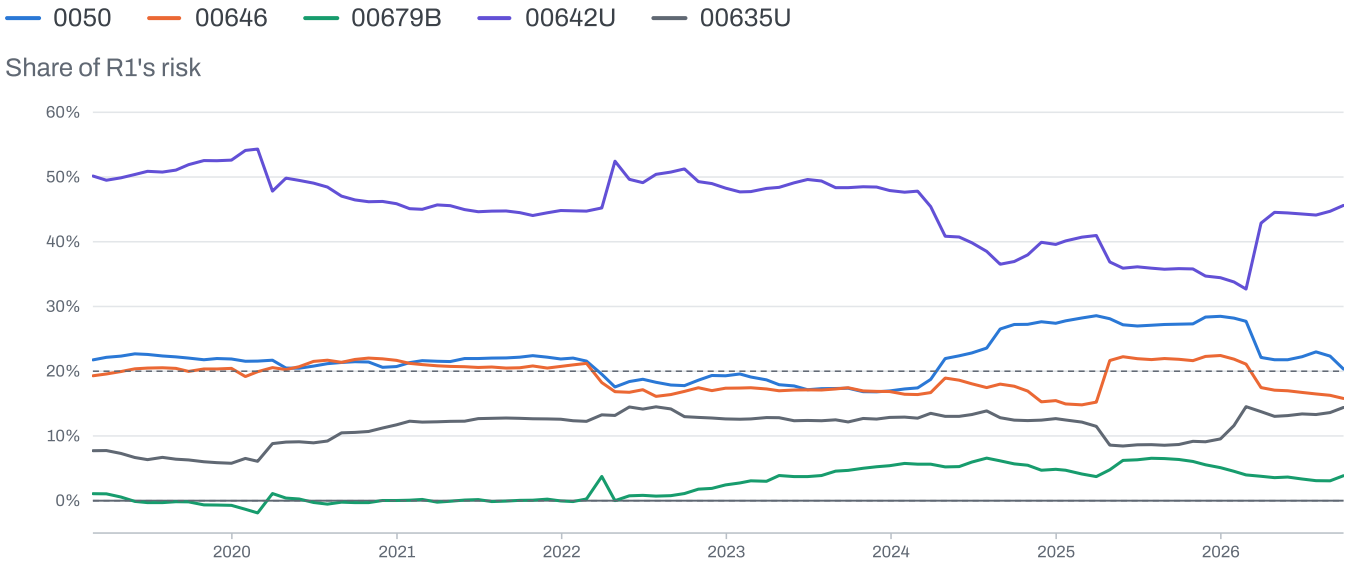
SOLUTION

There R1's s_p is 8.13%. 00679B correlates −0.067 with the book and has $\beta_{i,p} = -0.0944$, so its MRC is $-0.0944 \times 8.13\% = -0.77\%$ and its share $0.2 \times (-0.0944) = -1.89\%$. 00642U carries 54.3% of the risk. The switch lowers s_p to 7.07%: the money leaves the largest contribution and enters a negative one.

00679B's share was negative in 22 of the 36 windows from May 2019 to April 2022 (Exhibit 7); the last, through 29 April 2022, was −0.0023%.

Exhibit 7 repeats the split in the 92 windows that the rules read. 00642U's share ran from 32.7% to 54.3%, and 0050's from 16.8% to 28.6%. Each share describes its own two years, not the next.

Exhibit 7: R1's risk shares in its 92 trailing windows



Note: one point per month-end signal from 27 February 2019, each window holding the trailing 504 joint dates, annualized with the full sample's A. 00635U is grey; dashed, 0 and 20%. 00642U's lowest share is in the window through 26 February 2026. 00679B's last negative share, −0.0023% through 29 April 2022, is too small to see. The first window, from 13 February 2017, gives 21.7%, 19.3%, 1.1%, 7.7% and 50.2%. A stability check; LM30 measures the first window's error. Source: as Exhibit 1.

PITFALL **A negative share is not a hedge at any size**

00679B's share of −1.89% says that a little more of it lowers the volatility in that window. Its MRC rises with its weight, so a large enough position adds risk.

KNOWLEDGE CHECK 6

(a) 00642U carries 45.9% of R1's risk on the full sample and 32.7% to 54.3% across the 92 windows. Is 45.9% a forecast? (b) In the window through 27 February 2020, would selling some 00679B lower R1's risk?

ANSWER

(a) No. Each share describes its own sample, Exhibit 7 is a stability check, and LM30 measures the first window's error.

(b) No. In that window 00679B's MRC is −0.77% and the other MRCs are positive, so selling a little of it into any of them, or into cash, raises the volatility.

8 SYSTEMATIC AND RESIDUAL RISK

- [ ] split a portfolio's variance into systematic and residual parts with market betas

LM23's market model, fitted by (23.1), splits each sector's return into a part that moves with the market and a residual:

$$R_{i,t} = \alpha_i + \beta_i R_{\mathrm{mkt},t} + \varepsilon_{i,t}.$$

A book with weights $\mathbf{w}$ then has the beta $\mathbf{w}^\top \boldsymbol{\beta}$ and the residual $\sum_i w_i \varepsilon_{i,t}$. If the errors are uncorrelated with $R_{\mathrm{mkt},t}$, (16.4) gives

$$\begin{aligned} \sigma_p^2 &= (\mathbf{w}^\top \boldsymbol{\beta})^2 \sigma_{\mathrm{mkt}}^2 + \mathbf{w}^\top \Sigma_{\mathrm{res}} \mathbf{w}, \\ \mathrm{RC}_i &= \frac{w_i \beta_i (\mathbf{w}^\top \boldsymbol{\beta})\, \sigma_{\mathrm{mkt}}^2 + w_i (\Sigma_{\mathrm{res}} \mathbf{w})_i}{\sigma_p}. \end{aligned} \tag{27.8}$$

In sample (27.8) is exact when each β_i is the OLS slope with an intercept, because the normal equations make each residual series' sample covariance with $R_{\mathrm{mkt},t}$ zero (Problem 9). The first term is the **systematic** part, the second the **residual** part, and both split by fund.

U28 holds price indices, without dividends, and only their covariances are used. Under (23.8)'s assumption that dividends accrue evenly, the dividend gap moves the intercepts, not the covariances.

EXAMPLE 10

An equal-weight U28 book

Split the variance of LM10's equal-weight book of the 28 sector price indices, with the TAIEX price index as the market (Exhibit 8).

SOLUTION

The sectors' betas run from 0.507 to 1.207 (LM23), and the book's is $\mathbf{w}^\top \boldsymbol{\beta} = 0.807$. To three decimals, so that the shares sum, the systematic part is 83.426% of the variance, the R-squared of the book on the TAIEX price index. The residual part, 16.574%, is 4.037% from residual variances and 12.537% from residual covariances. These use unrounded inputs; the printed 0.807, 17.82% and 15.75% give 83.4%.

Setting the residual covariances to 0, as a single-index model does, gives 14.73% instead of 15.75%. The off-diagonal part is positive, consistent with the residuals' mean correlation of 0.112 (LM23), although a mean does not fix a weighted sum's sign. Semiconductor's share is $4.46\% - 0.64\% = 3.82\%$: its residual covaries negatively with the book's.

Exhibit 8: The U28 book's systematic and residual risk

Quantity	Value
The book's volatility	15.75%
TAIEX price index's volatility	17.82%
The book's beta, $\mathbf{w}^\top \boldsymbol{\beta}$	0.807
Systematic share of the variance (= R-squared)	83.426%
Residual share of the variance	16.574%
from the residual variances	4.037%
from the residual covariances	12.537%
Systematic and residual parts as volatilities	14.4%, 6.4%
Volatility with residual covariances set to 0	14.73%
Semiconductor's risk share: systematic and residual parts	4.46% − 0.64% = 3.82%
Largest risk share: GlassCeramic	5.21%
Smallest risk share: CommunicationsInternet	1.99%
Equal share, 1/28	3.57%

Note: U28, 4,382 daily simple price returns (no dividends) of 28 TWSE sector indices and the TAIEX price index, 2 December 2008 to 8 October 2026, $A = 245.4794$. OLS with an intercept, divisor T; exact in sample. Descriptive. Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026, with copied closes repaired from `TaiwanStockEvery5SecondsIndex` as in LM23. Computations: `data/lm27_numbers.py`.

Estimating a factor model's covariance, and the cost of ignoring residual covariances, is LM30's subject.

導讀 **系統性風險與殘差風險**

市場模型（market model）把每個類股的報酬拆成跟著大盤動的部分和自己的殘差（residual）。帳戶的 beta 是各類股 beta 的加權平均，帳戶變異數於是分成兩塊：beta 平方乘上大盤變異數是系統性風險（systematic risk），其餘來自殘差的共變異數矩陣（covariance matrix）。用最小平方法（least squares）加截距（intercept）估計時，殘差和大盤的樣本共變異數剛好是 0，所以這個變異數分解（variance decomposition）在樣本內完全成立。單一指標模型（single index model）假設殘差彼此不相關，但等權重類股帳戶的殘差平均正相關，略去它們會低估波動率：14.73%，實際是 15.75%。

KNOWLEDGE CHECK 7

Why does the systematic share of the U28 book's variance, 83.4%, equal the R-squared of the book's regression on the TAIEX price index?

ANSWER

OLS is linear in the response, so the book's slope is $\mathbf{w}^\top \boldsymbol{\beta}$. By (12.4) its R-squared is the explained variance over the total, $(\mathbf{w}^\top \boldsymbol{\beta})^2 \sigma^2_{\text{mkt}} / s_p^2$, the first term of (27.8) over s_p^2.

9 PORTFOLIO ACCOUNTS IN PYTHON

- ☐ compute risk contributions and the wealth recursion in NumPy, and check them against their identities

The lab's files hold E5's frames in long form, one row per date and fund, and the block below works from them.

PYTHON

```python
import numpy as np
import polars as pl

funds = ["0050", "00646", "00679B", "00635U", "00642U"]
est = pl.read_csv("data/e5_estimation.csv",
                  schema_overrides={"fund": pl.String})
X = (est.pivot(on="fund", index="date", values="R")
     .sort("date").select(funds).to_numpy())
A = 2362 * 365.25 / 3551                        # 242.951
S = A * np.cov(X, rowvar=False, ddof=0)         # divisor T
w = np.full(5, 0.2)
sp = np.sqrt(w @ S @ w)                         # 0.11493
rc = w * (S @ w) / sp                           # (27.6)
assert abs(rc.sum() - sp) < 1e-15               # Euler's identity
wf = pl.read_csv("data/e5_wealth.csv",
                 schema_overrides={"fund": pl.String})
d = pl.col("date").str.to_date()
wk = (wf.group_by("fund", d.dt.iso_year().alias("y"),
                  d.dt.week().alias("k"))
      .agg((pl.col("R") + 1).product() - 1)   # weekly returns
      .sort("fund", "y", "k"))   # 502 weeks; Section 4 keeps 498

def post_trade(W_pre, drift, f, tax, k=0.001425):
    W = W_pre                                   # (27.2) by iteration
    while True:
        x = W * f - W_pre * drift
        new = W_pre - k * np.abs(x).sum() - tax @ np.clip(-x, 0, None)
        if abs(new - W) < 1e-15 * W_pre:
            return new
        W = new
```

`np.cov` divides by $T-1$ unless given `ddof=0`. `select(funds)` fixes the column order, and the sort fixes the order of the groups, which `group_by` does not keep.

> **IN PYTHON** **Companion lab**
>
> The Polars-first notebook `LM27_lab.ipynb` rebuilds E5's two frames from the raw closes, the drift, R1's and R2's trades, the correlations by frequency, the risk splits and U28's factor split. It solves the computational practice problems and ends with R2's risk shares in the 92 windows.
>
> The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

`np.cov(X, rowvar=False)` divides by $T - 1$. What does that do to R1's full-sample 11.493% on 2,362 returns?

ANSWER

Every volatility rises by the factor $\sqrt{T/(T-1)} = 1.000212$, so R1's 11.493% becomes 11.495%.

SUMMARY

- A book's return uses the previous close's weights, (27.1); between trades the weights drift.
- Costs come out of wealth, so the post-trade wealth solves (27.2), which has one root.
- One-way turnover is half the 1-norm; a trade costs about 0.285% of it plus taxes, (27.3).
- Daily joint statistics use E5's 2,362 joint dates, wealth its 2,368 sessions; daily correlations can mislead either way.
- A book's mean and variance are $\mathbf{w}^\top \boldsymbol{\mu}$ and $\mathbf{w}^\top \Sigma \mathbf{w}$, (27.4); the ratio (27.7) is at least 1 for a long-only book.
- Euler's contributions (27.5)–(27.6) add up to σ_p and price a switch; a negative one hedges only at the margin.
- With market betas the variance splits exactly in sample into systematic and residual parts, (27.8).

PRACTICE PROBLEMS

Costs are a commission of 0.1425% on each side and a tax of 0.1% on sales of every E5 fund except 00679B.

1. Two funds are held 50/50 over a period in which they return +10% and −5%, with no trade. Fund 1's weight at the end is *closest* to:

   A. 0.55

   B. 0.537

   C. 0.50

2. A book's pre-trade weights are 0.30, 0.20, 0.15, 0.15 and 0.20 in E5's order, and it trades back to a fifth each. By (27.3), the cost as a share of pre-trade wealth is *closest* to:

   A. 0.0285%

   B. 0.0670%

   C. 0.0385%

3. Two funds are bought 50/50 and held. Their growth factors over the period are 1.40 and 0.80. The book's return is *closest* to:

   A. 10.00%

   B. 5.83%

   C. 5.67%

4. Three assets have volatilities 20%, 10% and 30%, correlations $\rho_{12} = 0.2$, $\rho_{13} = 0.6$ and $\rho_{23} = -0.2$, and weights 0.4, 0.4 and 0.2 (constructed). Asset 1's risk share by (27.6) is *closest* to:

   A. 36.2%

   B. 44.4%

   C. 56.1%

5. In a window, a fund held long ($w_i > 0$) has a negative risk contribution to the book. Which statement always holds?

   A. Moving a small weight into it from a fund with a positive MRC lowers the book's volatility.

   B. Raising its weight lowers the volatility at any size.

   C. Selling it lowers the volatility.

6. E5's monthly correlation of 0050 with 00646 is 0.5488 on 116 calendar months (descriptive). Its normal-theory standard error, to four decimals, is *closest* to:

   A. 0.0060

   B. 0.0649

   C. 0.0928

7. For $\sigma_p > 0$, derive $\partial\sigma_p/\partial w_i = (\Sigma\mathbf{w})_i/\sigma_p$ from (10.8), then (27.6) and $\sum_i w_i\beta_{i,p} = 1$. Show that fund i's Euler share of the variance is its variance term plus half of each covariance term that involves it.

8. Derive the solution of (27.2) when the signs of the trades are known, then (27.3)'s approximation and its bound. Check both on Example 3, whose exact cost is 0.0310028% of pre-trade wealth and approximation 0.0310231%.

9. Show that (27.8) is exact in sample when each β_i is the OLS slope with an intercept, and split RC_i into its two parts. If an equal-weight book's residual covariances average positive, what sign has the error from dropping them?

10. *(Python)* With Problem 4's inputs and annual means of 8%, 4% and 10% (constructed), compute μ_p, σ_p, the ratio (27.7), the risk contributions and the risk shares. Predict from the MRCs the change in σ_p when 5 points move from asset 3 to asset 2, and compute it by (10.9).

11. *(Python)* Build E5's two frames from the lab's raw files, and check Example 4's figures and Exhibit 4's correlations. Why can a daily correlation mislead here, and which of Exhibit 4's five pairs have daily–monthly gaps of two or more monthly normal-theory standard errors?

12. *(Python)* Run (27.2) for R1, fixed in advance, from cash of 1 at the close of 4 March 2019, trading at the lab's trade dates to R1's targets. Print the largest gap from the lab's net wealth path, then Exhibit 3's trade and R1's shares in the 92 windows.

SOLUTIONS

1. **B** is correct: per unit of start wealth the funds grow to 0.55 and 0.475, so fund 1's weight is $0.55/1.025 = 0.537$. A forgets the book's growth; C is the rebalanced weight.

2. **C** is correct: $\mathrm{TO}_t = \frac{1}{2}(0.10 + 0 + 0.05 + 0.05 + 0) = 0.10$, and only 0050's sale of 0.10 is taxed, so the cost is $0.285\% \times 0.10 + 0.1\% \times 0.10 = 0.0385\%$; the exact root also gives 0.0385%. A leaves out the tax, and B charges commission on twice the turnover.

3. **A** is correct: by (1.13) the growth factor is $0.5 \times 1.40 + 0.5 \times 0.80 = 1.10$, a return of 10.00%. B averages the log returns, $\sqrt{1.40 \times 0.80} - 1 = 5.83\%$, which understates a long-only book by (1.14). C reads that average log return, 0.0567, as a simple return.

4. **C** is correct: with $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$, the first entry of $\Sigma\mathbf{w}$ and the variance are

$$\begin{aligned}(\Sigma\mathbf{w})_1 &= 0.016 + 0.0016 + 0.0072 \\ &= 0.0248, \\ \sigma_p^2 &= 0.4 \times 0.0248 + 0.4 \times 0.0044 \\ &\quad + 0.2 \times 0.03 = 0.01768.\end{aligned}$$

   So the share is $0.4 \times 0.0248/0.01768 = 56.1\%$. A is the variance term, $0.4^2 \times 0.04/0.01768 = 36.2\%$, and B the stand-alone split, $0.08/0.18 = 44.4\%$.

5. **A** is correct: $w_i > 0$ and $\mathrm{RC}_i < 0$ give $\mathrm{MRC}_i < 0$, so moving a small weight from fund j changes σ_p by about that weight times $\mathrm{MRC}_i - \mathrm{MRC}_j < 0$. B fails, because $(\Sigma\mathbf{w})_i/\sigma_p = \rho_{i,p}\sigma_i$ gives

$$\begin{aligned}\frac{\partial\,\mathrm{MRC}_i}{\partial w_i} &= \frac{\sigma_i^2}{\sigma_p} - \frac{(\Sigma\mathbf{w})_i^2}{\sigma_p^3} \\ &= \frac{\sigma_i^2(1 - \rho_{i,p}^2)}{\sigma_p} \geq 0,\end{aligned}$$

   so MRC_i never falls and tends to $\sigma_i > 0$ as w_i grows; it turns positive at a large enough weight. C fails: selling a small weight into cash changes σ_p by about minus that weight times MRC_i, a rise.

6. **B** is correct: $(1 - 0.5488^2)/\sqrt{116} = 0.6988/10.770 = 0.0649$. A divides by T instead of $\sqrt{T}$, and C leaves out $1 - \hat{\rho}^2$. The plug-in standard error of LM20's Problem 6 is 0.081.

7. By (10.8), $\partial\sigma_p^2/\partial w_i = 2(\Sigma\mathbf{w})_i$, and the chain rule on $\sigma_p = \sqrt{\sigma_p^2}$ gives $(\Sigma\mathbf{w})_i/\sigma_p$. Then $\sum_i w_i\,\mathrm{MRC}_i = \mathbf{w}^\top\Sigma\mathbf{w}/\sigma_p = \sigma_p$, and $\mathrm{RC}_i/\sigma_p = w_i(\Sigma\mathbf{w})_i/\sigma_p^2 = w_i\beta_{i,p}$, whose sum is 1. Finally,

$$w_i(\Sigma\mathbf{w})_i = w_i^2\sigma_i^2 + \textstyle\sum_{j\neq i} w_i w_j \sigma_{ij},$$

and each $w_i w_j \sigma_{ij}$ is half of the term $2w_iw_j\sigma_{ij}$ of σ_p^2.

8. Sum separately over the funds bought (b) and sold (s), drop the date and write w_i for w_i^{drift}. Then

$$\begin{aligned} W = W^{\mathrm{pre}} &- 0.1425\%\textstyle\sum_{\mathrm{b}}(W f_i - W^{\mathrm{pre}} w_i) \\ &- 0.1425\%\textstyle\sum_{\mathrm{s}}(W^{\mathrm{pre}} w_i - W f_i) \\ &- \textstyle\sum_{\mathrm{s}} \mathrm{tax}_i (W^{\mathrm{pre}} w_i - W f_i) \end{aligned}$$

is linear in W, so

$$\frac{W}{W^{\mathrm{pre}}} = \frac{1 + 0.1425\%\sum_{\mathrm{b}} w_i}{1 + 0.1425\%\sum_{\mathrm{b}} f_i} \cdot \frac{-\sum_{\mathrm{s}}(0.1425\% + \mathrm{tax}_i)\, w_i}{-\sum_{\mathrm{s}}(0.1425\% + \mathrm{tax}_i)\, f_i}.$$

On Example 3 the four buys' drifted weights sum to 0.71942 and 00642U's is 0.28058, so the formula gives $W/W^{\mathrm{pre}} = 1.0003448/1.000655 = 0.9996900$, a cost of 0.03100%. Replacing W by W^{pre} inside the cost gives (27.3). The two trade vectors differ by $(W^{\mathrm{pre}} - W)\mathbf{f}$, of 1-norm cost $\times \|\mathbf{f}\|_1$, and the cost is Lipschitz with constant $0.1425\% + \max_i \mathrm{tax}_i$ (Deep Dive 1): that is the bound. On the same trade the gap, 0.0000203% of pre-trade wealth, is within the bound $0.0310028\% \times 0.2425\% = 0.0000752\%$.

9. With an intercept, the normal equations give each residual series a zero mean and a zero sample covariance with $R_{\mathrm{mkt},t}$. So $\hat\Sigma = \sigma_{\mathrm{mkt}}^2\boldsymbol{\beta}\boldsymbol{\beta}^\top + \Sigma_{\mathrm{res}}$ exactly, and (27.8) follows from $\mathbf{w}^\top\hat\Sigma\mathbf{w}$ and $(\hat\Sigma\mathbf{w})_i$. With equal weights the dropped part is N^{-2} times the sum of the off-diagonal residual covariances, positive when they average positive. Dropping them understates the variance: 14.73% against 15.75% in Example 10.

10. The lab gives the mean

$$\begin{aligned} \mu_p = 0.4 \times 8\% + 0.4 \times 4\% \\ + 0.2 \times 10\% = 6.80\%, \end{aligned}$$

a volatility of 13.30% and the ratio $18\%/13.30\% = 1.35$. The risk contributions are 7.461, 1.324 and 4.512 points, summing to 13.297, and the shares 56.1%, 10.0% and 33.9%. The MRCs, 18.65%, 3.31% and 22.56%, predict $0.05 \times (3.31\% - 22.56\%) = -0.96$ points; by (10.9) σ_p falls by 0.89 points, to 12.41%.

11. The lab reproduces every figure. 00646's and 00679B's 13:30 prices follow the previous New York session, while 0050 also moves with Taipei's own trading. The futures funds are valued on European and US futures (Section 4). Of the five pairs, only 00679B–00642U's daily–monthly gap reaches two monthly normal-theory standard errors, 2.2 (Example 5).

12. The lab's recursion matches the shared path to below 10^{-12} at every date, and it prints Exhibit 3 and Exhibit 7.

GLOSSARY

Term	中文	Meaning
Book	帳戶	The positions held together; between trades their weights drift, (27.1).
Drift of weights	權重漂移	The change of a book's weights between trades, (27.1).
Buy-and-hold	買進持有	A book that keeps its holdings, so its weights drift with prices.
Rebalancing	再平衡	A trade back to target weights.
Self-financing	自我融資	Costs are paid out of wealth; no money comes in or goes out, (27.2).
Wealth recursion	—	(27.2): the pre-trade wealth, the trade and the post-trade wealth.
One-way turnover	週轉率	$\mathrm{TO}_t = \frac{1}{2}\lVert \mathbf{f}_t - \mathbf{w}_t^{\mathrm{drift}} \rVert_1$, (27.3).
Transaction cost	交易成本	Commission on both sides and tax on sales, paid at the trade.
Estimation frame	—	E5's 2,362 dates on which every fund has a one-session return.
Wealth frame	—	E5's 2,368 sessions; a halted fund earns 0 until it trades again.
Asynchronous closes	非同步收盤	Prices that reflect different sessions abroad at the same close.
Diversification ratio	—	$\mathbf{w}^\top \mathbf{s} / s_p$, at least 1 for a long-only book, (27.7).
Beta to the portfolio	—	$\beta_{i,p} = \mathrm{Cov}(R_i, R_p)/\sigma_p^2$, (27.5).
Marginal risk contribution	邊際風險貢獻	$\mathrm{MRC}_i = \partial \sigma_p / \partial w_i = (\Sigma \mathbf{w})_i / \sigma_p$, (27.6).
Risk contribution	風險貢獻	$\mathrm{RC}_i = w_i \, \mathrm{MRC}_i$; the contributions sum to σ_p.
Risk share	風險份額	$\mathrm{RC}_i / \sigma_p = w_i \beta_{i,p}$.
Euler's identity	—	$\sum_i w_i \, \partial \sigma_p / \partial w_i = \sigma_p$, since σ_p is homogeneous of degree 1.
Euler's theorem	Euler 定理	A differentiable h with $h(a\mathbf{x}) = a\,h(\mathbf{x})$ for every $a > 0$ has $\sum_i x_i \, \partial h / \partial x_i = h(\mathbf{x})$ (Deep Dive 2).
Systematic risk	系統性風險	$(\mathbf{w}^\top \boldsymbol{\beta})^2 \sigma_{\mathrm{mkt}}^2$, the market part of the variance, (27.8).
Residual risk	殘差風險	$\mathbf{w}^\top \Sigma_{\mathrm{res}} \mathbf{w}$, the rest of the variance, (27.8).

Term	中文	Meaning
Single index model	單一指標模型	A market model whose residuals are taken as uncorrelated.

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